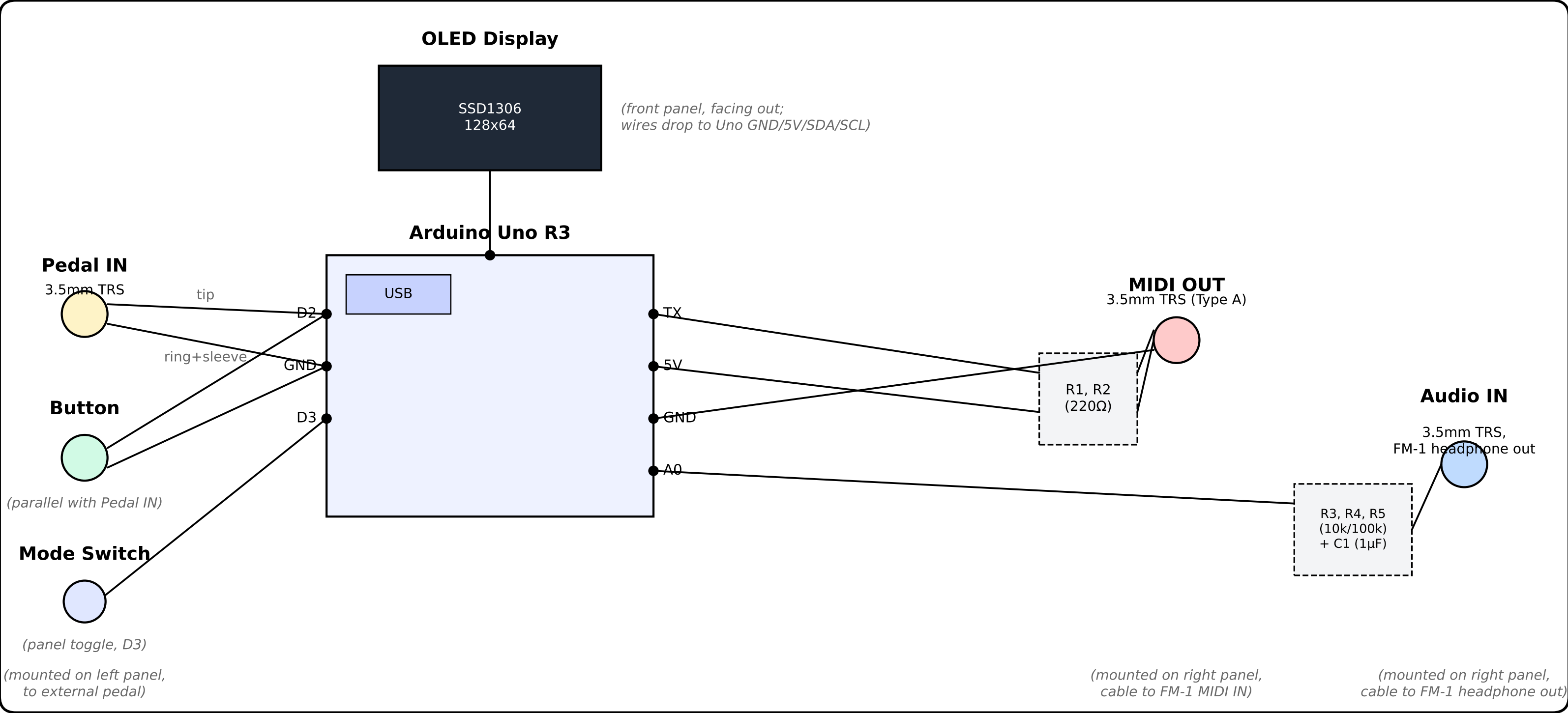


FM-1 Sustain Footswitch — Layout / Assembly Diagram

Top-down view of enclosure: pedal jack + built-in button + mode switch (left panel), Arduino Uno (center), MIDI out + audio in jacks (right panel), OLED (front panel)

project box (top cover removed)



Notes:

- Resistors R1/R2 can live on a small offcut of perfboard, or be soldered directly in-line on the wire runs -- exact placement isn't critical
- See the schematic for the electrical connections (which wire goes through which resistor) -- this diagram is for physical placement only
- Button and Pedal IN jack are wired in parallel to the same D2/GND nodes -- either one alone triggers sustain, so the pedal can be left unplugged
- Leave slack on the USB cable path -- the board still needs to be reachable for re-flashing (disconnect MIDI OUT jack's TX lead first)
- Mode Switch, OLED, and Audio IN are all optional additions -- the board works exactly as the original sustain-only build without them
- OLED needs 4 wires to the Uno (GND, VCC/5V, SDA->A4, SCL->A5) in addition to the single SDA/SCL point shown here for clarity